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MILITARY SPECIFICATION

EXTINGUISHERS, FIRE, CARBON DIOXIDE, 5-POUND AND 10-POUND CAPACITY, FIXED TYPE (NONSHATTERABLE CYLINDERS)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers carbon dioxide (CO_2) fire extinguishers, 5 and 10 pound capacity, with flood valve, anti-recoil device, and nonshatterable cylinder.

1.2 Classification. Extinguishers shall be classified as follows (see 6.2):

Size 5 - Five-pound capacity with 1/2 inch flood valve.
Size 10 - Ten-pound capacity with 1 inch flood valve.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: DRSTA-GSS, Warren, MI 48090, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC 4210

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SPECIFICATIONS
FEDERAL

BB-C-101	- Carbon Dioxide (CO_2): Technical and U.S.P.
BB-N-411	- Nitrogen, Technical.
QQ-C-576	- Copper Flat Products with Slit, Slit and Edge-rolled, Sheared, Sawed, or Machined Edges (Plate, Bar, Sheet, and Strip).
QQ-S-781	- Strapping, Steel, and Seals.
TT-E-489	- Enamel, Alkyd, Gloss (for Exterior and Interior Surfaces).
TT-I-1795	- Ink, Marking, Stencil, Opaque (Porous and Non Porous Surfaces).
PPP-B-601	- Boxes, Wood, Cleated Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-640	- Boxes, Fiberboard, Corrugated, Triple-wall.

MILITARY

MIL-T-704	- Treatment and Painting of Materiel.
MIL-E-52835	- Enamel, Modified Alkyd, Camouflage, Lusterless.

STANDARDS
FEDERAL

FED-STD-H28	- Screw Thread Standards for Federal Services.
FED-STD-595	- Color.

MILITARY

MIL-STD-101	- Color Code for Pipelines and for Compressed Gas Cylinders.
MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of U.S. Military Property.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.
MIL-STD-45662	- Calibration Systems Requirements.

2.1.2 Other Government documents, drawings, and publications. The following documents, drawings, and publications form a part of this specification to the extent specified herein.

DEPARTMENT OF TRANSPORTATION (DoT)

49 CFR 1-199	- Department of Transportation Rules and Regulations.

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(Applications for copies should be addressed to Department of Transportation, Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

DRAWINGS
ARMY

7714476	- Cylinder Assembly, Commercial (Size 10).
8386355	- Valve Control, Carbon Dioxide.
10946814	- Cylinder, Compressed Gas (Size 5).
10947623	- Valve Control, Assembly.
11669713	- Safety Disc.
11669714	- Safety Disc Gasket.
11669715	- Safety Disc Nut.
12257169	- Cylinder Assembly (Size 10).

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM B117	- Standard Method of Salt Spray (Fog) Testing.

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA. 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. Unless otherwise specified, the contractor shall furnish extinguishers which shall be subjected to first article inspection (see 4.3). First article inspection samples, properly marked with identifying information shall be representative of the unit to be furnished to the Government. All subsequent extinguishers delivered to the Government shall conform to these samples in all of their pertinent physical and performance characteristics.

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3.2 Materials. Materials shall be as specified herein and in referenced specifications, standards, and drawings. Material shall be free of defects which adversely affect performance or serviceability of finished product (see 6.3).

3.3 Construction.

3.3.1 Cylinder. The cylinders shall be of seamless steel or aluminum construction, shall not be wire wound and shall have a service pressure of 1,800 pounds per square inch (psi). Cylinders shall be stable when standing with the longitudinal axis in a vertical position, valve end up.

3.3.1.1 Steel. Steel cylinders shall conform to DoT Code of Federal Regulations, Title 49, Specification 3AA, Section 178.37, and shall conform to the applicable drawings.

3.3.1.2 Aluminum. Aluminum cylinders shall conform to DoT Code of Federal Regulations, Title 49, Specification 3AL, Section 178.46, with dimensions as specified by the acquisitioning activity.

3.3.2 Flood valve. The flood valve, conforming to Drawing 7714476 for the size 10 extinguisher and Drawing 10946814 for the size 5 extinguisher, shall be actuated by an external control valve. The flood valve shall include a safety release mechanism, a siphon tube, and an anti-recoil device.

3.3.3 Safety release mechanism. The safety release mechanism shall consist of:

a. Safety Disc conforming to Drawing 11669713. 1/
b. Gasket, Safety Disc conforming to Drawing 11669714. 2/
c. Nut, Safety Disc conforming to Drawing 11669715. 3/

1/ Optional Safety Disc - Same as 11669713 except material shall be covered with 14 Karat rolled gold.
2/ Optional Gasket, Safety Disc - Same as 11669714 except material shall conform to QQ-C-576, Copper, cold rolled or soft annealed.
3/ Cutting edge to be sharp (no radius or chamfer), diameter .281 ID (inside diameter) ± .002.

3.3.4 Siphon tube. The flood valve siphon tube shall be fabricated of non-ferrous, corrosion resistant material of sufficient strength to withstand continuous usage without breaking. The siphon tube shall extend inside the cylinder to a depth not more than 1/2 inch from the cylinder bottom. The tube shall connect to the valve body by a threaded, flared, preformed shoulder or compression fitting. Soldered (sweated) connections shall not be permitted.

3.3.5 Threaded parts. Screw threads, of the form, number per inch, and class specified on the applicable drawing or military standard, shall be in accordance with the applicable sections of FED-STD-H28.

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3.3.6 Dimensions.

3.3.6.1 Size 5 extinguishers. The extinguishers shall conform to Drawing 10946814 and shall have a capacity of not less than 204 cubic inches. The weight of the extinguisher, fully assembled and charged with carbon dioxide (CO_2), shall be not more than 21 pounds.

3.3.6.2 Size 10 extinguishers. The extinguishers shall conform to Drawing 7714476 and shall have a capacity of not less than 408 cubic inches. The weight of the extinguisher, fully assembled and charged with CO_2, shall be not more than 45 pounds.

3.4 Fire extinguisher charge.

3.4.1 Standard charge. The extinguishers shall be charged with a standard charge of CO_2 conforming to BB-C-101, Grade B, and with an oil content of no more than 10 parts per million (ppm) by weight. The weight of the standard charge shall be equal to the extinguisher rating for each size and shall not be more than 68 percent of the water capacity (by weight) of the cylinder.

3.4.2 High temperature charge. The high temperature charge shall be 90 percent of the standard charge.

3.4.3 Low temperature charge. Dry (water pumped) nitrogen conforming to BB-N-411 shall be charged into the extinguishers to a pressure of 200 psi at 70°F, or the corresponding pressure for other temperatures as specified in table I. The extinguishers shall then be placed on a scale, and the weight of liquefied carbon dioxide specified in table II shall be added.

TABLE I. Nitrogen pressure.

Ambient temperature: degrees Fahrenheit	Pressure: psi
30	185
40	188
50	192
60	196
70	200
80	204
90	207
100	211
110	215
120	219

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TABLE II. Weight of carbon dioxide - low temperature.

Size number	Weight: pounds
5	3.50
10	7.35

3.5 Performance.

3.5.1 Flood valve. When diassembled from the extinguisher and with the safety release mechanism removed, the flood valve shall withstand 6,000 pounds per square inch (psi) without permanent deformation or leakage and without impairing function (see 4.5.1).

3.5.2 Safety release mechanism. The safety disc shall rupture at not less than 2,650 psi and not more than 3,000 psi, and shall relieve all internal pressure (see 4.5.2).

3.5.3 Extinguisher.

3.5.3.1 Normal temperature leakage. The extinguisher, assembled and with standard charge, shall not perceptively leak and shall retain contents and pressure during a 4 hour exposure to temperatures between 90°F and 110°F (see 4.5.3.1 and 4.5.3.2).

3.5.3.2 High temperature leakage. The extinguisher, assembled and with extreme high temperature charge, shall not perceptively leak and shall retain contents and pressure during an 8 hour exposure to temperatures between 110°F and 145°F (see 4.5.3.3).

3.5.3.3 Low temperature leakage. The extinguisher, assembled and with extreme low temperature charge, shall not perceptively leak and shall retain contents and pressure during a 48 hour exposure to temperatures between minus 40°F and minus 25°F (see 4.5.3.4).

3.5.3.4 Operating effort. The effort required to operate the flood valve of a properly filled extinguisher, in 70°F ambient air temperature, shall not exceed the values specified in table III (see 4.5.4.1 and 4.5.4.2).

TABLE III. Operating effort.

Extinguisher	Maximum operating effort
Size 5	22 pounds
Size 10	32 pounds

3.5.3.5 Discharge characteristics. Extinguishers, assembled and with the charge specified herein, with the longitudinal axis in the vertical posotion and with the valve end up, shall discharge 97 percent of their contents within the time limit specified in table IV (see 4.5.5).

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TABLE IV. Speed of discharge.

Temperature	Size 5	Size 10 1/	Size 10 2/
+70°F	6 seconds	3 seconds	6 seconds
-40°F	8 seconds	4 seconds	8 seconds

1/ Without diffuser tip in place.
2/ With diffuser tip in place.
3/ Diffuser tip must be removable from the outlet port of the size 10 bottle.

3.5.3.6 Anti-recoil. An anti-recoil device, as shown on Drawings 10946184 and 12257169 for the size 5 and size 10 extinguishers, respectively, shall be provided to absorb all reaction to discharge of internal pressure from a fully charged extinguisher. The devices shall be a permanent (non-removable) part of the discharge system (see 4.5.6).

3.5.3.7 Corrosion resistance. Extinguishers shall withstand 200 hours of salt spray exposure without impairing function, and without impairing manipulation of fittings and connections with common tools (see 4.5.7).

3.5.3.8 Vibration resistance. Extinguishers, assembled and charged as specified herein, shall withstand vibration testing specified in MIL-STD-810, method 514.2, procedure VIII, except that the vibration frequency and amplitude shall be as specified in table V. The extinguisher shall exhibit no weight loss due to testing (see 4.5.8).

TABLE V. Vibration frequency and amplitude.

Frequency: cycles per second (Hz)	Amplitude: inches of gravity (g)
5.0-9.9	0.5 inch double amplitude
9.9-180	2.5g
180-500	2.5g to 1.5g

3.5.3.9 Physical shock resistance. Extinguishers, assembled and charged as specified herein, shall exhibit no weight loss due to shock testing as specified in MIL-STD-810, method 516.2, procedure I (see 4.5.9).

3.5.3.10 Nonshatterability.

a. 3AA, 4130X steel cylinders over 4 inches in diameter, with wall stress at the minimum specified test pressure under 70,000 psi, and 3AA, 4130X steel cylinders under 4 inches in diameter, with wall stress at the minimum specified test pressure under 60,000 psi shall be certified as nonshatterable and shall be permanently marked "NONSHAT".

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b. 3AA, 4130X steel cylinders under 4 inches in diameter, with wall stress at the minimum specified test pressure over 60,000 psi and aluminum cylinders shall be subjected to and pass the gunfire test specified in 4.5.10 to become acceptable as nonshatterable (see 4.5.10 and 6.4).

3.6 Finish.

3.6.1 Size 5. Portions of extinguishers normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, type A. Final coat shall be in accordance with TT-E-489, color red, gloss no. 11105 of FED-STD-595, as specified in MIL-STD-101. When specified (see 6.2), color shall be forest green in accordance with MIL-E-52835.

3.6.2 Size 10. The cylinder shall be cleaned, treated, and painted in accordance with MIL-T-704, type A. Final coat shall be in accordance with TT-E-489, color red, gloss no. 11105 of FED-STD-595, as specified in MIL-STD-101.

3.7 Marking.

3.7.1 Cylinder. Each cylinder shall be plainly and permanently marked on the shoulder of the cylinder by metal stamping in characters not less than 1/4 inch high, with the following data in accordance with the requirements of MIL-STD-130:

One side.
a. DoT specification and service pressure and "SPUN" when spun cylinders are supplied.
b. The symbol of the Department of the Army - "DA".
c. Manufacturer's serial number - letters.
d. Manufacturer's name, trademark, or DoT registered symbol.
e. "NONSHAT".

Other side. Opposite the above marking and as near the neck as possible.

a. Acquisition document number.
b. Original hydrostatic test date (month and year).

In addition, each cylinder shall be plainly and permanently marked on the sidewall by silk screening or by using black ink conforming to TT-I-1795, the following in characters not less than 1/2 inch high (order of information optional):

a. CARBON DIOXIDE
b. Wt. Empty Incl. Valve . . . lbs.
c. Charge to . . . lbs. CO_2 1/
d. Rigid siphon tube.

1/ 5 lbs. or 10 lbs. as applicable.

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3.7.2 Valve. The valve shall bear the following cast or stamped on one side of the valve body (blank spaces shall be filled in prior to delivery):

Full_____pounds
Empty_____pounds

"Full" shall be the weight of the "Empty" cylinder plus the weight of the CO_2 charge. "Empty" shall be the weight of the uncharged cylinder plus the weight of the valve and siphon tube.

3.8 Instruction and record cards. Each extinguisher shall be accompanied by a card bearing operating instructions, and one bearing recharging instructions. Further, each extinguisher shall be equipped with two cards suitable for recording weight and recharge data. Instruction cards shall be subject to approval by the contracting officer.

3.9 Workmanship. After final assembly, extinguishers shall be cleaned of loose, spattered, or excess solder, metal chips, and other foreign material. Burrs and sharp edges, as well as resin flash that may crumble, shall be removed. Workmanship shall be of such quality as to insure that extinguishers furnished under this specificatin are free of defects that compromise, limit, or reduce performance in the intended use (see 6.1).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform or witness any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection equipment. Unless otherwise specified in the contract, the supplier is responsible for the provision and maintenance of all inspection equipment necessary to assure that supplies and services conform to contract requirements. Commercial, modified commercial, or supplier designed inspection equipment or measuring set-ups must be capable of repetitive measurements to an accuracy of 10 percent of the component tolerance. Calibration of inspection equipment shall be in accordance with MIL-STD-45662.

4.1.2 Materials and construction. To determine conformance to 3.2 through 3.4.3, inspection and material certification records shall be maintained by the contractor. Records shall be subject to review by the Government and shall include date, part, or characteristic identification, inspection results, and disposition of lot (accepted or rejected). Corrective action taken on noted defects shall be subject to approval by the Government.

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4.2 Classification of inspection. Inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspections (see 4.4).
   1. Examination (see 4.4.2).
   2. Acceptance tests (see 4.4.3).

4.3 First article inspection. Unless otherwise specified (see 6.2), the Government shall select four extinguishers produced under the production contract (see 6.2) for first article inspection. First article samples shall be examined as specified in table VI and tested as specified in table VII. Approval of the first article sample by the Government shall not relieve the contractor of his obligation to supply extinguishers that are fully representative of those inspected as a first article sample. Any changes or deviation of the production units from the first article sample shall be subject to the approval of the contracting officer.

4.3.1 First article inspection failure. Failure of any first article sample to pass specified examinations or tests shall be cause for refusal to grant first article approval until corrective action by the contractor has been approved by the Government.

4.4 Quality conformance inspection.

4.4.1 Sampling.

4.4.1.1 Lot formation. An lot shall consist of all the extinguishers of one type, from an identifiable production period, from one manufacturer, subjected at one time for inspection.

4.4.1.2 Sampling for examination. Samples for quality conformance examination shall be selected in accordance with MIL-STD-105.

4.4.2 Quality conformance examination.

4.4.2.1 Acceptable quality level. Each sample selected in accordance with 4.4.1.2 shall be examined to determine conformance to the following acceptable quality levels (AQL) on the basis of percent defective.

Classification	AQL
Major defect	2.5
Minor defect	4.0

4.4.2.2 Classification of defects. For examination purposes, defects shall be classified as listed in table VI.

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TABLE VI. Classification of defects.

Category	Defect	Method of examination
Critical	None	
Major	AQL 2.5% Defective	
101	Flood valve not as specified (see 3.3.2).	Visual
102	Siphon tube not as specified (see 3.3.4).	Visual
103	Improper threads (see 3.3.5).	Visual/Gage
104	Incorrect dimensions affecting interchangeability (see 3.3.6).	Visual/Gage
105	Improper weight (see 3.3.6).	Gage
106	Extinguishing agent (charge) pressure not as specified (see 3.4).	Gage
Minor	AQL 4.0% Defective	
201	Unstable when standing (see 3.3.1).	Visual
202	Incorrect dimensions not affecting interchangeability (see 3.3.6).	Visual/Gage
203	Improper finish (see 3.6).	Visual
204	Improper marking (see 3.7).	Visual
205	Instruction cards missing or non-conforming (see 3.8).	Visual
206	Poor workmanship (see 3.9).	Visual

4.4.3 Acceptance tests. Each extinguisher shall be subjected to test for leakage at normal temperature as specified in table VII.

4.4.3.1 Acceptance test failure. Failure of any sample to pass acceptance tests shall be cause for Government refusal to accept extinguishers until corrective action taken by the contractor has been approved by the Government.

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4.5 Methods of inspection.

TABLE VII. Classification of tests.

Title	Requirements	First article	Acceptance tests
Flood valve	3.5.1	4.5.1	
Safety release mechanism	3.5.2	4.5.2	
Leakage	3.5.3	4.5.3	
Normal temperature	3.5.3.1	4.5.3.1	4.5.3.1
Normal temperature (optional)	3.5.3.1	4.5.3.2	4.5.3.2
High temperature	3.5.3.2	4.5.3.3	
Low temperature	3.5.3.3	4.5.3.4	
Operating effort	3.5.3.4	4.5.4	
Size 5	3.5.3.4	4.5.4.1	
Size 10	3.5.3.4	4.5.4.2	
Discharge characteristics	3.5.3.5	4.5.5	
Anti-recoil device	3.5.3.6	4.5.6	
Corrosion resistance	3.5.3.7	4.5.7	
Vibration resistance	3.5.3.8	4.5.8	
Physical shock resistance	3.5.3.9	4.5.9	
Nonshatterability	3.5.3.10.b	4.5.10	

4.5.1 Flood valve test. To determine conformance to 3.5.1, the flood valve shall be removed from the extinguisher, and the safety release mechanism removed from the flood valve. The flood valve shall be installed in the apparatus to apply internal pressure by the hydrostatic pump. Admit pressure through the inlet. All outlets shall be plugged. Pressure shall be raised to 6,000 psi. Any permanent deformity of the valve which results shall be cause for rejection.

4.5.2 Safety release test. To determine conformance to 3.5.2, the flood valve shall be removed from the extinguisher. With the safety release in place, the valve shall be installed in the apparatus to apply pressure by the hydrostatic pump and to admit pressure through the inlet. All outlets shall be plugged except the safety release. Care shall be taken to avoid supporting the safety disc. Pressure shall be raised to 2,500 psi and then increased at a rate of 100 psi until rupture occurs. Rupture at below 2,650 psi, or above 3,000 psi, shall be cause for rejection.

4.5.3 Leakage test. To determine conformance to 3.5.3, the extinguishers shall undergo the procedures specified in 4.5.3.1 (or 4.5.3.2), 4.5.3.3, and 4.5.3.4. After testing, remove all traces of water from the valve and the safety release mechanism.

4.5.3.1 Normal temperature. To determine conformance to 3.5.3.1, the extinguisher, assembled and charged as specified herein, shall be sumberged in water maintained at a temperature of 100°F $\pm$ 10°F for not less than four hours. Throughout submersion, the extinguisher shall be completely covered by a bell jar filled with water in order to trap and collect gas which may escape. Any trace of escaped gas shall constitute failure of this test.

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4.5.3.2 Normal temperature (alternate test method). To determine conformance to 3.5.3.1, an alternate test method utilizing a leak detector capable of detecting carbon dioxide leak rates of less than one ounce per year may be used in lieu of 4.5.3.1.

4.5.3.3 High temperature. To determine conformance to 3.5.3.2, the extinguisher, assembled and charged as specified herein, shall be submerged in water maintained at a temperature of 145°F for not less than eight hours. Throughout submersion, the extinguisher shall be completely covered by a bell jar filled with water in order to trap and collect gas which may escape. Any trace of escaped gas shall consitute failure of this test.

4.5.3.4 Low temperature. To determine conformance to 3.5.3.3, the charged extinguisher shall be weighed. The extinguisher shall be subjected to a temperature of minus 40°F for a period of 48 hours, and subsequently reweighed. Any loss of weight shall constitute failure of this test.

4.5.4 Operating effort test. To determine conformance to 3.5.3.4, the extinguisher, assembled and filled as specified herein, shall undergo procedure specified in 4.5.4.1 or 4.5.4.2, as applicable.

4.5.4.1 Size 5. Attach control valve 10947623, or equal, to the flood valve. Attach a length of 1/16 inch diameter, 7x7, steel cable (wire rope) to the rotating cam in the control valve of a fully charged extinguisher. Hook the free end to an accurate spring scale. Pull the scale slowly until discharge occurs. Observe the reading on the scale to determine the effort required. An effort greater than 22 pounds shall constitute failure of this test.

4.5.4.2 Size 10. Attach control valve 8386355, or equal, to the flood valve. Attach a length of 1/16 inch diameter, 7x7, steel cable (wire rope) to the rotating cam in the control valve. Hook the free end to an accurate spring scale. Pull the scale slowly until discharge occurs. Observe the reading on the scale to determine the effort required. An effort greater than 32 pounds shall constitute failure of this test.

4.5.5 Discharge characteristics test. To determine conformance to 3.5.3.5, the extinguisher shall be subjected to following procedure: 1/

a. In 70°F ambient air temperature, the extinguisher shall be weighed before filling, and the weight recorded.
b. The extinguisher shall be charged (filled) with extinguishing agenct (charge) as specified herein.
c. The extinguisher with charge shall be weighed, and the weight recorded. The difference between reading a and reading c shall be determined, and shall be considered the weight of the contents. The difference (weight of contents) shall be recorded.
d. The extinguisher shall be subjected to ambient air temperatures as follows:

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(1)	Low temperature test	-40°F	48 hours minimum
(2)	Normal temperature test	70°F	until temperature is stabilized
(3)	High temperature test	145°F	until temperature is stabilized

e. After each test in d, the extinguisher shall be subjected to the following routine:

(1) The extinguisher shall be weighed, and the weight recorded. Any loss of weight when compared with c shall constitute failure of this test.
(2) An acceptable extinguisher shall be placed in a vertical position, and arrangements made to observe the time, in seconds, during discharge.
(3) The extinguisher shall be discharged and the weight observed (see table IV). The weight shall be recorded.
(4) The difference between reading c and reading e.(3) shall be not less than 97 percent of difference (weight of contents) determined in c. Difference less than 97 percent shall constitute failure of this test.

1/ Testing specified in 4.5.3 and 4.5.5 may be conducted concurrently.

4.5.6 Anti-recoil test. To determine conformance to 3.5.3.6, place a fully charged extinguisher on a smooth horizontal surface with the longitudinal axis of the extinguisher parallel to the smooth surface. Open the discharge valve fully. Any movement of the extinguisher from its original position, resulting from recoil, shall constitute failure of this test. Movement resulting from the force applied to open the valve shall be ignored.

4.5.7 Corrosion resistance test. To determine conformance to 3.5.3.7, the extinguisher shall undergo the tests specified in ASTM B117 for 200 hours.

4.5.8 Vibration resistance test. To determine conformance to 3.5.3.8, the extinguisher, assembled and charged as specified herein, shall be tested as specified in MIL-STD-810, method 514, procedure VIII, substituting the vibration frequency and amplitude specified in table V. The tests shall operate for a total of 120 minutes each, in the plane of each axis, including 20 minutes dwell time at each of 4 points of resonance. The extinguisher shall be weighed before and after the vibration test to assure that no loss of charge has occurred during the test. Any loss of weight shall constitute failure of this test.

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4.5.9 Physical shock resistance test. To determine conformance to 3.5.3.9, the extinguisher, assembled and charged as specified herein, shall be tested as specified in MIL-STD-810, method 516, procedure I, for 18 half-sine wave shock pulses (three in each direction on each axis) of a peak amplitude of 40g ± 4g, with the time duration of 18 ± 3 milliseconds. Further, the extinguisher shall undergo 18 pulses (three in each direction on each axis) of a peak amplitude of 60g ± 6g, with time duration of 3 ± 0.1 milliseconds. The extinguisher shall be weighed before and after the shock test. Any loss of weight shall constitute failure of this test.

4.5.10 Gunfire test. To determine conformance to 3.5.3.10.b, two cylinders shall be charged to the rated pressure plus or minus 5 percent, using a nonliquified gas. The cylinders shall be placed behind a suitable steel barricade. The cylinder shall be in such a position that a bullet passing through a hole in the barricade, strikes the cylinder at right angles to the longitudinal centerline within 1 inch of the longitudinal centerline and near the vertical center of the cylinder. The cylinder temperature at the time of the test shall be between 50° and 100°F. An armor piercing projectile 0.50 caliber in size shall be fired at the cylinder. The 0.50 caliber projectile shall strike the cylinder at a velocity of 2,800 feet per second, plus or minus 100 feet per second. The projectile shall strike the cylinder straight on (not tumbled). A cylinder shall be considered as having failed this test if the cylinder breaks into more than two pieces; provided, however, that pieces smaller than 2 inches in diameter coming from the areas (centering on the perforation and 4 inches in diameter) on the cylinder adjacent to the point of entry and exit of the projectile will not be counted. Cylinders, representative samples of which have passed this test, shall be permanently marked to indicate this fact. The term "NONSHAT" shall be permanently marked on the shoulder of the cylinder as specified in 3.7.1.

4.6 Inspection of packaging (Army use). Materials and processes involved in preparation for delivery shall be inspected to determine conformance to requirements of 5.1.1. Deviation from specified requirements may cause the Government to reject shipments associated with the non-conforming items until the contractor's plan to correct deficiencies and prevent recurrence has been approved by the Government.

4.7 Inspection of packaging (other agencies).

4.7.1 Quality conformance inspection of pack.

4.7.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.7.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105, level II.

4.7.1.3 Examination. Samples selected in accordance with 4.7.1.2 shall be examined for the defects specified in table VIII. AQL shall be 2.5 percent defective for major defects.

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TABLE VIII. Classification of packaging defects.

Categories	Defects	Method of inspection
Major		
101	Containers not as specified. Each incorrect container shall be considered one defect.	Visual
102	Extinguisher not blocked and cushioned within the box as specified.	Visual
103	Box closure not method V as specified.	Visual
104	Strapping not as specified.	Visual
105	Extinguishers of different sizes packed in the same container.	Visual
106	Marking illegible, incorrect, incomplete.	Visual

5. PACKAGING

5.1 Army use.

5.1.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking shall be in accordance with the applicable packaging data sheet, furnished at the time of invitation for bid, for the level of packaging, and level of packing, specified by the acquisitioning activity (see 6.2).

5.2 Other activities.

5.2.1 Preservation. Preservation shall be level A, or commercial, as specified (see 6.2).

5.2.1.1 Level A. Each extinguisher, with instruction cards, shall be preserved in a close-fitting box conforming to PPP-B-636, W6c for size 5 extinguisher, and W5c for size 10 extinguisher. Blocking and cushioning shall be provided to prevent free movement within the box. Box closure shall be in accordance with the appendix to the box specification, method V.

5.2.1.2 Commercial. Each extinguisher, with instruction cards, shall be preserved in accordance with MIL-STD-1188.

5.2.2 Packing. Packing shall be level A, level B, or commercial, as specified (see 6.2).

5.2.2.1 Level A. Extinguishers of like size, preserved as specified in 5.1, shall be packed together in close-fitting boxes conforming to PPP-B-601, overseas type, style optional. Strapping shall conform to QQ-S-781, class 1, type I or IV, size as applicable. Unless otherwise specified (see 6.2), strapping shall be finish B.

5.2.2.2 Level B. Packing shall be as specified in 5.2.2.1 for level A, except that boxes shall conform to PPP-B-601, domestic type, style optional, or PPP-B-640, class 2, style optional, and strapping shall be finish A.

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5.2.2.3 Commercial. Extinguishers of like size, preserved as specified in 5.1, shall be packed in accordance with MIL-STD-1188.

5.2.3 Marking. Marking shall be level A, level B, or commercial, as specified (see 6.2).

5.2.3.1 Military. Military marking, level A or level B, shall conform to MIL-STD-129.

5.2.3.2 Commercial. Commercial marking shall include item name, national stock number, quantity, unit of issue, contract number, and appropriate address markings. Marking shall be applied with waterproof material, be legible, and be of a color which contrasts definely with color surface on which it is applied. Marking shall conform to MIL-STD-1188.

6. NOTES

6.1 Intended use. Fire extinguishers are intended for use in combating class B fires, which involve oils, greases, gasoline, alcohol, and other volatile liquids; and class C fires, which involve electrical or electrically energized, equipment.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Size of extinguisher required (see 1.2).
c. If first article samples shall not be furnished (see 3.1 and 4.3).
d. Color, when other than that specified is required (see 3.6.1).
e. If supplier shall be responsible for the provision and maintenance of all inspection equipment (see 4.1.1).
f. Selection of applicable level of packaging, and level of packing (see 5.1.1, 5.2.1, or 5.2.2).
g. Type of strapping finish required (see 5.2.2.1).
h. Level of marking required (see 5.2.3).

6.3 Recycled materials. The use of recycled materials which meet the requirements of the applicable material specifications without jeopardizing the intended use of the item shall be encouraged (see 3.2).

6.4 Nonshatterability. Historically, Government agencies have gunfire tested high pressure cylinders to evaluate their nonshatterability and fragmentation properties. In applications where military personnel are in confined quarters, prime interest is to limit framentation and control release of the gas in a cylinder when it is pierced or burst. When ground support cylinders over 4 inches in diameter are considered, fragmentation in gunfire test is held to not more than two pieces. Extensive testing has verified that cylinders over 4 inches in diameter made to the limits of the DoT 3AA specification meet this requirement and have been marked "NONSHAT" by the fabricator for a number of years. Cylinders in aircraft service, where the controlled release of the contained gas is more critical, tearing about the projectile apertures is limited to 3 inches from the hole center. Government and industry have verified that this quality of nonshatterability

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is most directly related to the average wall stress in cylinder design. The DoT 3AA specification allows an average wall stress of 70,000 psi maximum. Independent investigators have arrived at wall stress limits from 50,000 to 60,000 psi as valid maximums for satisfactory "NONSHAT" characteristics. For acquisition in accordance with this specification, cylinders under 4 inches in diameter, fabricated in accordance with DoT 3AA specification, with a 60,000 psi maximum wall stress, will be accepted and shall be permanently marked "NONSHAT". Cylinders under 4 inches in diameter with maximum wall stress greater than 60,000 psi will pass actual gunfire tests before qualifying to be permanently marked "NONSHAT".

6.5 Changes from previous issue. Asterisks to identify changes with respect to the previous issue are not used in this revision due to the extensiveness of the changes.

Custodian:
Army - AT
Air Force - 99

Review activities:
Army - ME, MD
Air Force - 84
DLA - CS

Preparing activity:
Army - AT

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3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
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5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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